

Performance Testing

Project: Rising Water – AI-Based Flood Prediction System

1. Objective

Evaluate the performance, reliability, and responsiveness of the flood prediction application.

Test ID	Scenario	Expected Result	Status
PT-01	Load application	Home page opens successfully	Pass
PT-02	Enter valid inputs	Accept all 20 values	Pass
PT-03	Predict flood risk	Probability and risk displayed	Pass
PT-04	Invalid input	Validation message shown	Pass
PT-05	Multiple predictions	Application remains stable	Pass
PT-06	Response time	Prediction within a few seconds	Pass

2. Performance Metrics

Metric	Observation
Application Availability	100% during testing
Prediction Accuracy	Based on trained ML model
Average Response Time	< 3 seconds
UI Responsiveness	Good
Overall Result	Successful

3. Conclusion

The Rising Water application performed successfully during testing. The system accepted valid inputs, generated predictions quickly, and displayed flood probability with the appropriate risk level.